Southeast Tech · · Computer Information Systems

Data Analysis

CIS-137 30 O

Summer Term 2024-2025 School Year   Section 30 O   3.00 Credits   05/19/2025 to 07/25/2025
Modified 05/06/2025

Course Description

Students will learn how to find data from reliable sources, then preparte and visualize the data. Additional topics will include Integrating API's, Predictive Analysis, Linear Regression, Multiple Regression, JSON Files, Time Series Data and Types of Data Plots. Prerequisite: CSC 105, CIS 130

Outcomes

Technical Skills

- Dataframes:
  - Students will utilize dataframes for storing and analyzing data
- Visualization:
  - Students will create appropriate visualizations to support the trends and stories they discover in data
- Machine Learning
  - Students will utilize PyTorch to create a predictive model for image classification.

Critical Thinking

- Analysis:
  - Students will discover trends and stories emerging from data

Professionalism/Communication

- Submission requirements for the assignment are met. This includes video length, file formats, etc.
- Students will collect data from real world targets
- Students will find appropriate sources for data

Course Materials

Murach's Python for Data Analysis

Author: Scott McCoy
Publisher: Murach
ISBN: 978-1-943872-76-3
Price: ~$60 paperback, ~$55 ebook

Deliverables

Homework

Weekly assignments will be used to track attendance. They will be a completion grade, and are intended as student self assessments. The instructor will not make corrections to the homework. It is intended that students study the material provided until they have a good understanding of it.

Unit Project

There are 3 units in this course, each with a Unit Project. Unit Projects will be the assessments for this course, where you will create an application to demonstrate skills, and then present the application to prove your skills. Presentations may be recorded if not attending class on presentation day.

In addition to the presentation, your instructor may require an interview on the Unit Projects to answer follow-up questions and gauge your understanding of the material.

Within Unit Projects, students will be assessed on competencies, which will be used to determine their Project Competencies - Approaching, Proficient, and Mastered scores.

Evaluation Procedures and Grading

Criteria

Please Note: The intent of this grading system is for students to understand each topic in the course in order to pass. Completion of the homework and demonstrated mastery of the course topics is encouraged to raise your grade beyond a C, presuming that you are meeting proficiency in all of the competencies.

Type	Weight	Topic	Notes
Project Competencies - Approaching	40%		Student achieves at least a 2/4 on every project competency
Project Competencies - Proficient	30%		Student achieves at least a 3/4 on every project competency

Type	Weight	Topic	Notes
Project Competencies - Mastery	10%		Each 4/4 score on project competencies, students will receive partial credit towards this score. Example: a student achieving 4/4 on half of the competencies will receive 5% of their final grade weight from this.
Participation/Homework	20%		Homework assignments weighted equally will give partial credit towards this score.

Breakdown

This breakdown is a model/example, not a hard rule for the final grading. Final grading will depend on the grade weighting as defined in the criteria.

Grade	Range	Notes
A	90%-100%	For an A, it is expected that students attend every class session , complete every competency with a score of 3/4 or higher, complete all homework, and completing some competencies with a score of 4/4.
B	80%-89%	For a B, it is expected that students achieve a 3/4 on every competency and also complete most homework assignments, and possibly receive a few 4/4 scores on competencies.
C	70%-79%	C is the minimum passing grade for this course. To receive a C in this course, a student must complete every competency on each assessment with a 3/4 or higher.
D	60% - 69%	Students participating completing all homework but not passing every competency with a 3/4 or higher may expect to receive a D in this course, if they complete every competency with a 2/4.

Course Outline

When	Topic	Notes
Unit 1 Weeks 1-4	Intro to Data Analysis	• Python data types • Basic Python • JupyterNotebook • Dataframes • Manipulating columns and rows • Plotting/visualizing data • Collecting Data

When	Topic	Notes
Unit 2 Weeks 5-7	Advanced Analysis Skills	• Cleaning Data • Interpreting info about data • Pivot tables, aggregates, bins • Time Series data • Resampling • Rolling windows
Unit 3 Weeks 8-10	Predictive Analysis and Machine Learning	• Linear Regression • Training models • PyTorch • Google Colab • Neural Networks